

Лабораторная Работа №10. Настройка списков управления доступом (ACL)

Администрирование локальных сетей

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Освоить настройку прав доступа пользователей к ресурсам сети.

Подсоединение ноутбука

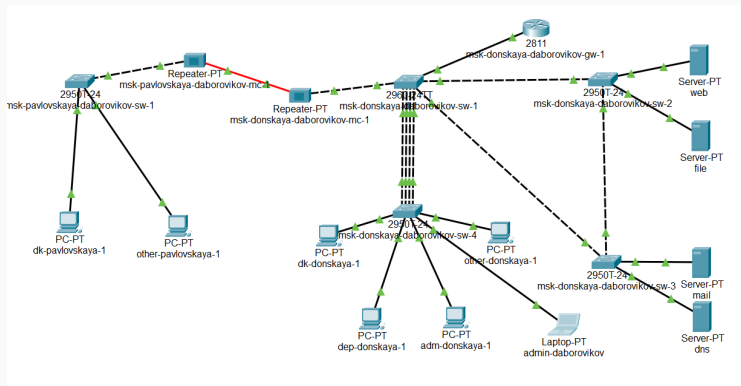


Figure 1: Подсоединение ноутбука к порту 24 коммутатора sw-4

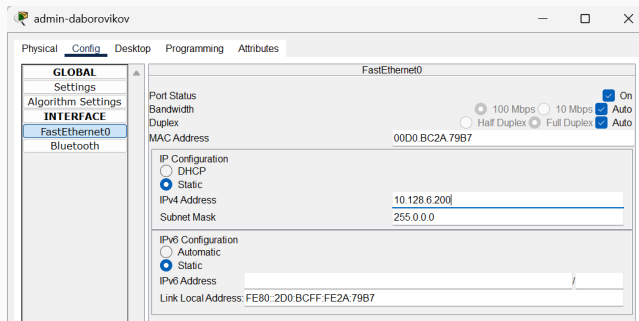
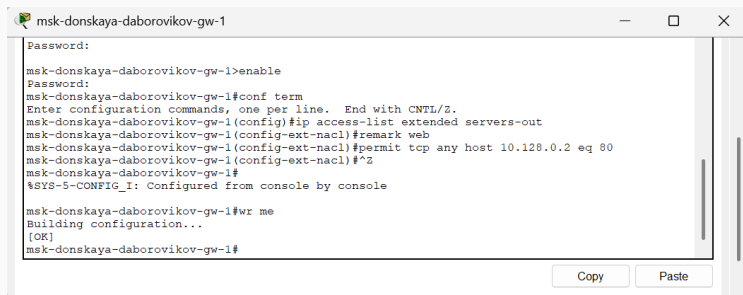


Figure 2: Присвоение оконечному устройству статического адреса 10.128.6.200, gateway-адреса 10.128.6.1 и адреса DNS-сервера 10.128.0.5

Настройка доступа к web-серверу



```
msk-donskaya-daborovikov-gw-1
Password:
msk-donskaya-daborovikov-gw-1>enable
Password:
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark web
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.2 eq 80
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-gw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#
```

Figure 3: Настройка доступа к web-серверу по порту tcp 80 (создан список контроля доступа с названием servers-out; указано, что ограничения предназначены для работы с web-сервером; дано разрешение доступа по протоколу TCP всем пользователям сети на доступ к web-серверу, имеющему адрес 10.128.0.2, через порт 80)

Список управления доступом к интерфейсу

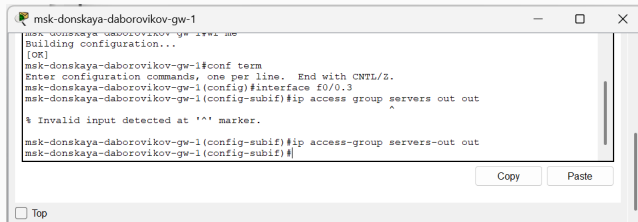
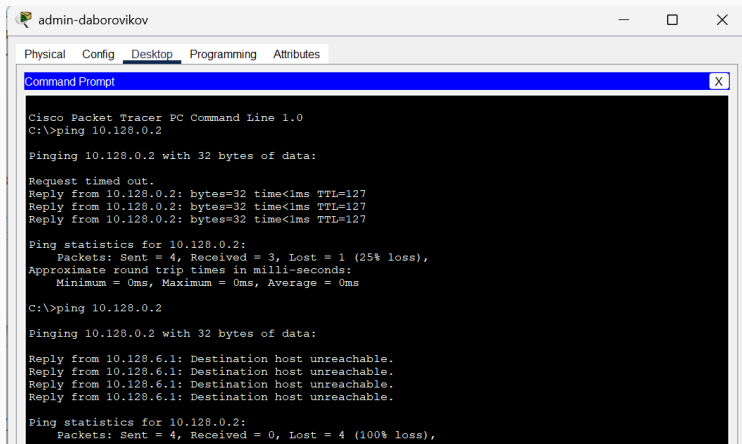


Figure 4: Добавление списка управления доступом к интерфейсу (к интерфейсу f0/0.3 подключается список прав доступа serversout и применяется к исходящему трафику)



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'admin-daborovikov'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.6.1: Destination host unreachable.
Reply from 10.128.6.1: Destination host unreachable.
Reply from 10.128.6.1: Destination host unreachable.
Reply from 10.128.6.1: Destination host unreachable.

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 5: ping

Проверка доступа к web-серверу через протокол HTTP

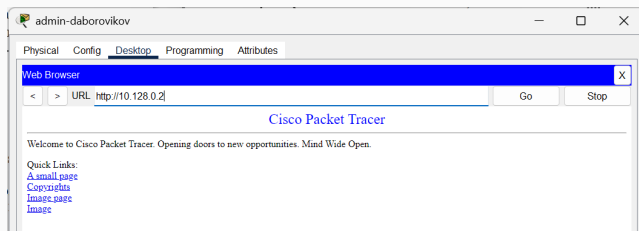
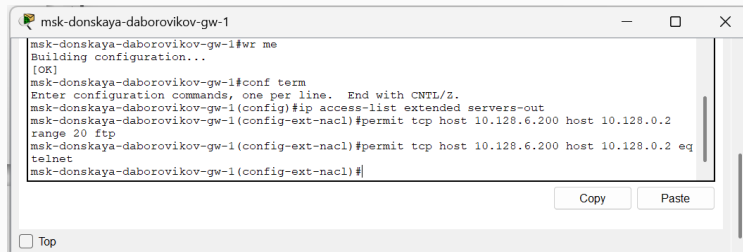


Figure 6: Проверка доступа к web-серверу через протокол HTTP



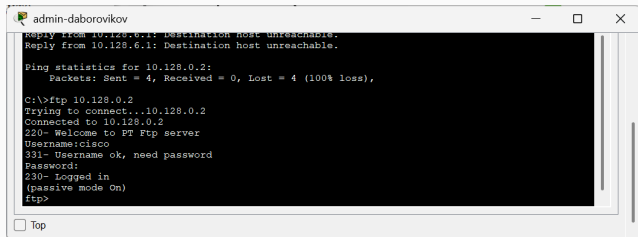
```
mzk-donskaya-daborovikov-gw-1
mzk-donskaya-daborovikov-gw-1#wr me
Building configuration...
[OK]
mzk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
mzk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp host 10.128.6.200 host 10.128.0.2
range 20 ftp
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp host 10.128.6.200 host 10.128.0.2 eq
telnet
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#
```

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Figure 7: Настройка дополнительного доступа для администратора по протоколам Telnet и FTP (в список контроля доступа servers-out добавлено правило, разрешающее устройству администратора с ip-адресом 10.128.6.200 доступ на web-сервер (10.128.0.2) по протоколам FTP и telnet)

Проверка доступа по протоколу FTP



```
admin-daborovikov
Reply from 10.128.6.1: Destination host unreachable.
Reply from 10.128.6.1: Destination host unreachable.

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
Connected to 10.128.0.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

Figure 8: Проверка доступа по протоколу FTP

Проверка доступа по протоколу FTP

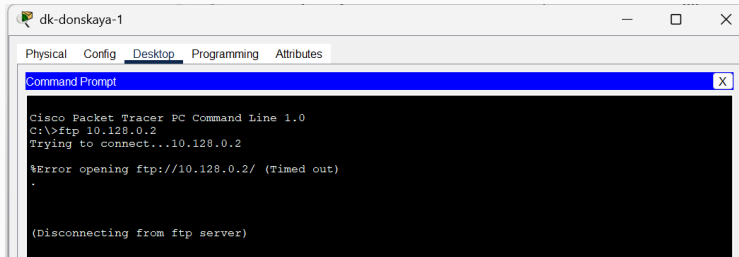
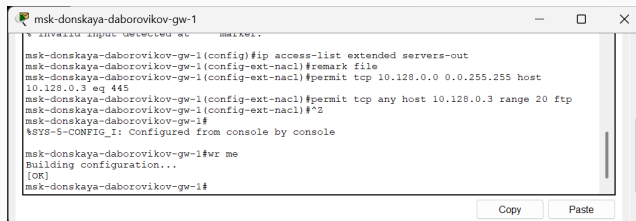


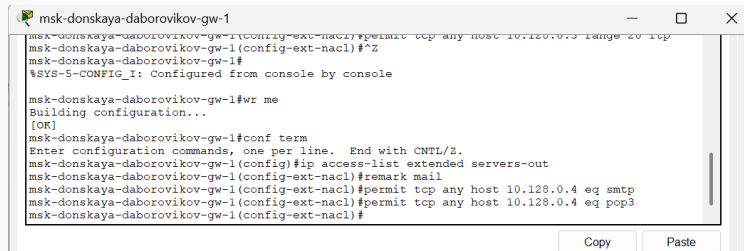
Figure 9: Проверка доступа по протоколу FTP

Настройка доступа к файловому серверу



```
msk-donskaya-daborovikov-gw-1
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark file
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp 10.128.0.0 0.0.255.255 host
10.128.0.3 eq 445
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.3 range 20 ftp
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-daborovikov-gw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#
```

Figure 10: Настройка доступа к файловому серверу (всем узлам внутренней сети (10.128.0.0) разрешён доступ по протоколу SMB к каталогам общего пользования; любым узлам разрешён доступ к file-серверу по протоколу FTP)

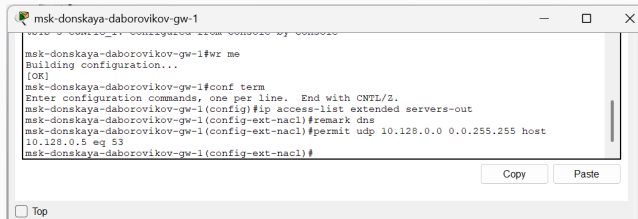


```
msh-donskaya-daborovikov-gw-1
msh-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.4 range 20 25
msh-donskaya-daborovikov-gw-1(config-ext-nacl)#^Z
msh-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msh-donskaya-daborovikov-gw-1#wr me
Building configuration...
[OK]
msh-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msh-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
msh-donskaya-daborovikov-gw-1(config-ext-nacl)#remark mail
msh-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.4 eq smtp
msh-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.4 eq pop3
msh-donskaya-daborovikov-gw-1(config-ext-nacl)#
```

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Figure 11: Настройка доступа к почтовому серверу (всем разрешён доступ к почтовому серверу по протоколам POP3 и SMTP)



```
msk-donskaya-daborovikov-gw-1
msk-donskaya-daborovikov-gw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark dns
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit udp 10.128.0.0 0.0.255.255 host
10.128.0.5 eq 53
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#
```

Figure 12: Настройка доступа к DNS-серверу (всем узлам внутренней сети разрешён доступ к DNS-серверу через UDP-порт 53)

Проверка доступности web-сервера (через браузер) по имени

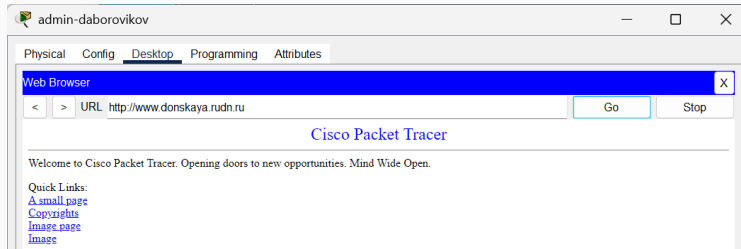
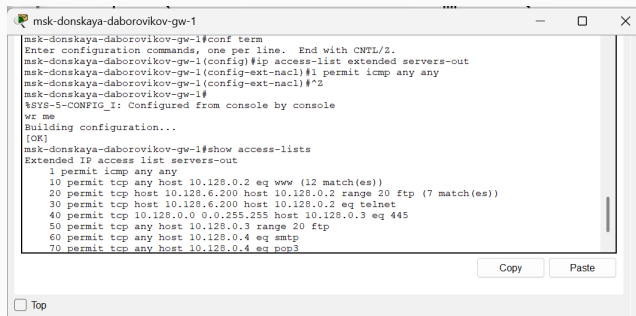


Figure 13: Проверка доступности web-сервера (через браузер) по имени

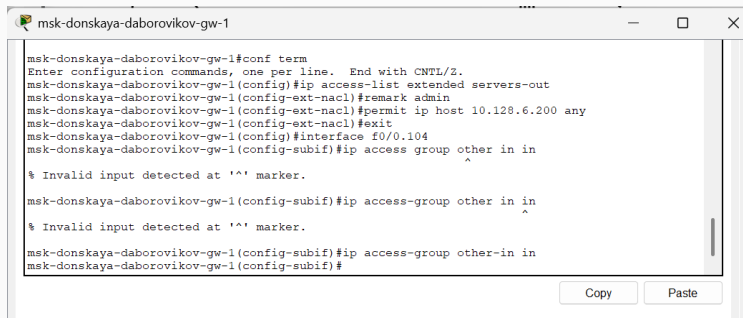
Разрешение icmp-запросов



```
msk-donskaya-daborovikov-gw-1
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#1 permit icmp any any
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#show access-lists
Extended IP access list servers-out
 1 permit icmp any any
10 permit tcp any host 10.128.0.2 eq www (12 match(es))
20 permit tcp host 10.128.6.200 host 10.128.0.2 range 20 ftp (7 match(es))
30 permit tcp host 10.128.6.200 host 10.128.0.2 eq telnet
40 permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445
50 permit tcp any host 10.128.0.3 range 20 ftp
60 permit tcp any host 10.128.0.4 eq smtp
70 permit tcp any host 10.128.0.4 eq pop3
```

Figure 14: Разрешение icmp-запросов (icmp-запрос добавляется в начало списка контроля доступа)

Настройка доступа для сети Other



```
msk-donskaya-daborovikov-gw-1
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark admin
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.128.6.200 any
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/0.104
msk-donskaya-daborovikov-gw-1(config-subif)#ip access group other in in
% Invalid input detected at '^' marker.
msk-donskaya-daborovikov-gw-1(config-subif)#ip access-group other in in
% Invalid input detected at '^' marker.
msk-donskaya-daborovikov-gw-1(config-subif)#ip access-group other-in in
msk-donskaya-daborovikov-gw-1(config-subif)#
```

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Figure 15: Настройка доступа для сети Other (разрешение устройству с адресом 10.128.6.200 на любые действия; подключение к интерфейсу f0/0.104 списка прав доступа other-in и применение к входящему трафику)

Настройка доступа администратора

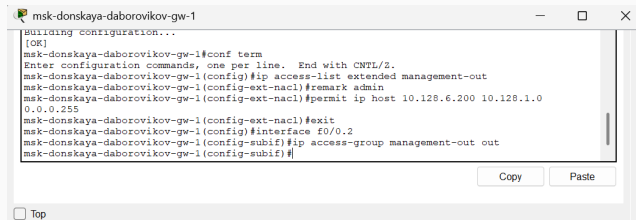
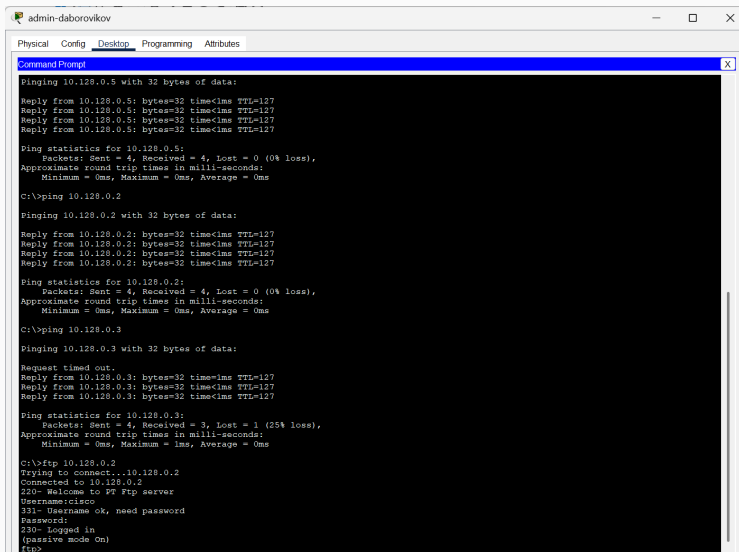


Figure 16: Настройка доступа администратора к сети сетевого оборудования (10.128.6.200 разрешён доступ к сети сетевого оборудования (10.128.1.0); к интерфейсу f0/0.2 подключён список прав доступа management-out и применено к исходящему трафику)

ping ftp



The screenshot shows a Windows Command Prompt window titled "admin-daborovikov". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" selected. The Command Prompt displays the following output:

```
Command Prompt
Pinging 10.128.0.5 with 32 bytes of data:
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.3

Pinging 10.128.0.3 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.3: bytes=32 time=1ms TTL=127
Reply from 10.128.0.3: bytes=32 time<1ms TTL=127
Reply from 10.128.0.3: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
Connected to 10.128.0.2
220- Welcome to PT Ftp server
Username:ricso
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

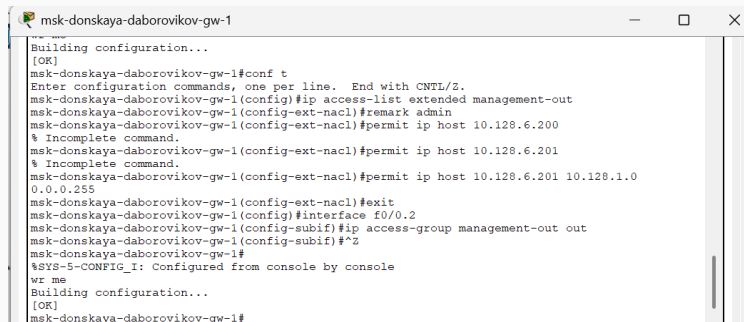
Figure 17: ping ftp

```
msk-donskaya-daborovikov-gw-1>en
Password:
msk-donskaya-daborovikov-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#configure terminal
^
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-out
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp host 10.128.6.201 host 10.128.0.2
range 20 ftp
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp host 10.128.6.201 host 10.128.0.2 eq
telnet
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-in
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended servers-in
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark admin
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.128.6.201 any
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/0.104
msk-donskaya-daborovikov-gw-1(config-subif)#ip access group other-in in
^
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-gw-1(config-subif)#ip access-group other-in in
```

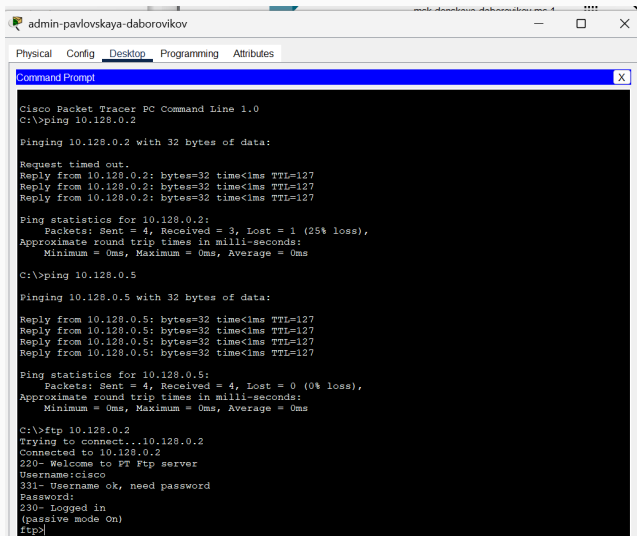
Figure 18: Разрешение администратору из сети Other на Павловской действия, аналогичные действиям администратора сети Other на Донской.



```
mzk-donskaya-daborovikov-gw-1
Building configuration...
[OK]
mzk-donskaya-daborovikov-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
mzk-donskaya-daborovikov-gw-1(config)#ip access-list extended management-out
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark admin
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.128.6.200
% Incomplete command.
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.128.6.201
% Incomplete command.
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.128.6.201 10.128.1.0
0.0.0.255
mzk-donskaya-daborovikov-gw-1(config-ext-nacl)#exit
mzk-donskaya-daborovikov-gw-1(config)#interface f0/0.2
mzk-donskaya-daborovikov-gw-1(config-subif)#ip access-group management-out out
mzk-donskaya-daborovikov-gw-1(config-subif)#^Z
mzk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
mzk-donskaya-daborovikov-gw-1#
```

Figure 19: Разрешение администратору из сети Other на Павловской действия, аналогичные действиям администратора сети Other на Донской.

Проверка Other на Павловской



```
admin-pavlovskaya-daborovikov
Physical Config Desktop Programming Attributes
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.5

Pinging 10.128.0.5 with 32 bytes of data:

Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
Connected to 10.128.0.2
220- Welcome to FT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

Figure 20: Проверка разрешений администратора из сети Other на Павловской

В ходе выполнения лабораторной работы я освоил настройку прав доступа пользователей к ресурсам сети.